Parametrization of Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise

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Abstract

In many signal processing applications, defining and finding peaks is an important part of the pipeline. Peak prediction can be a very challenging endeavor, especially when there is a lot of noise. In this work, the peak region is considered as a combination of a Gaussian trend with additive asymmetric Laplace noise, and the peak is modeled as the peak of the Gaussian trend.

We propose a criterion based on the weighted quantile which predicts the optimal number of observations necessary for the exact prediction of the peak's location. We analyze the effect of the signal-to-noise ratio on our prediction. We show that for small signal-to-noise ratio (a lot of noise), the prediction becomes inaccurate.

In this case, we propose a new idea called the entry time into the region of the peak and the exit time from the region of the peak. We demonstrate how to apply our new idea to real data and we analyze the effect of the choice of the objective function. As a numerical application, we consider the peak prediction of the daily infections during the first wave of COVID-19 in China, France, and Germany.

Keywords: Gaussian model, least absolute deviation, weighted median, entry time, region of the peak, daily infection.

Mathematics subject classification: 62M10, 60G35.

1 Introduction

Predicting peaks in time series data is crucial across various domains for proactive decision-making, risk management, resource allocation, and disaster preparedness. The ability to anticipate spikes in data can serve as an early warning system for events such as disease outbreaks in epidemiology, allowing public health officials to allocate resources and implement preventive measures effectively. In energy management, predicting peak electricity demand enables utility companies to adjust generation and distribution schedules, ensuring a reliable power supply

and preventing shortages. Financial analysts use peak prediction to manage investment risks by anticipating fluctuations in stock prices or market volatility. Additionally, environmental scientists rely on peak prediction to monitor pollution levels and natural disasters, facilitating disaster preparedness and response efforts. The examples justifying the usefulness of locating, measuring, and understanding these peaks are numerous.

Predicting peaks in time series data is a complex task due to several reasons. Firstly, the presence of noise and variability in the data makes it difficult to accurately identify and predict peaks. This noise can originate from various sources, such as measurement errors or random fluctuations in the data. Additionally, many real-world systems exhibit nonlinear dynamics, meaning that the relationship between variables is not strictly linear. Nonlinearities can lead to unexpected behaviors in the data, further complicating peak prediction. Moreover, time series data often contain complex patterns, such as multiple overlapping peaks or irregular intervals between peaks, making it challenging to develop accurate prediction models. In the literature, existing peak prediction methods often rely on specific assumptions about the data distribution or peak characteristics, which may not always hold true in practice, leading to inaccurate predictions (see for instance, **?** and **?**).

Popular forecasting models such as Autoregressive Integrated Moving Average (ARIMA) and Recurrent Neural Networks (RNN) often struggle to accurately define or forecast peak values in time series data. Despite their effectiveness in capturing general trends and patterns, these models may overlook or underestimate the occurrence of peaks due to their complexity and the inherent variability of peak events.

In the realm of peak prediction, there has been relatively little theoretical work compared to more generalized forecasting approaches. One notable contribution in this area is Tucker thesis **?**, which delved into the dynamics of trend, seasonality, and noise in time series data to predict peaks. Tucker research explored how the interplay between these components could be leveraged to identify and forecast peak events accurately. Inspired by previous studies **?**, Tucker applied a combination of general prediction techniques and specialized peak prediction methods. He utilized the Peaks function available in Scipy's Signal module, which offers tools for detecting and analyzing peak values in data. Tucker thesis represents a significant step forward in the field of peak prediction, offering insights into the complex dynamics underlying peak events and proposing practical methods for their prediction. By building on existing theoretical frameworks and leveraging advanced computational tools, Tucker work contributes to the growing body of research aimed at improving the accuracy and reliability of peak forecasting in time series data.

We adopt in this work the same approach as in **?** . We assume that the peak region is equal to $[T, h]$, where T represents the current date and h a specific time horizon. To model this hypothesis effectively, we consider our time series over the interval $[1, T]$ as a composite

of a Gaussian trend and noise. Our primary objective is to forecast the amplitude A, location L, and width S of the Gaussian trend based on observed data from the interval $[1, T]$. We introduce a criterion based on the weighted quantile technique, which enables us to determine the optimal number of observations required for accurately predicting the location L of the peak. Additionally, we conduct an analysis to assess the impact of the signal-to-noise ratio on our predictive accuracy. We show that in case of low signal-to-noise ratios, prediction accuracy diminishes. To address this challenge, we propose as in **?** a novel concept termed the "peak region entry and exit times".

The aim of our study is to extend the work of **?** to the quantile regression. First, we introduce our approach to selecting the minimization of the quantile regression and demonstrate, using simulated data, the number of observations required to accurately predict the location of the peak. This approach is based on the concept of the weighted quantile. Secondly, we establish that for high signal-to-noise ratios, the peak location prediction is correct. However, in cases of low `SNR` (indicating high noise levels), the accuracy of the prediction deteriorates. In these cases, we propose a new concept called "peak region entry time" and "peak region exit time". We illustrate this concept by parameterizing the entry time in the peak region and the exit time from the peak region by the tau-quantile when the data are the daily COVID-19 infections during the first wave in China **?**, France and Germany.

The remainder of this paper is organized as follows. In Section 2 we present our proposed model. More precisely, we consider Gaussian trend with additive generalized Laplace noise. Section **??** is dedicated to application results to forecast the peak incidence of daily COVID-19 infections during the initial wave in China, France, and Germany.

2 Quantile-Based Estimation Methodology

We consider the problem of estimating the amplitude, location, and spread of a unimodal signal corrupted by noise. The model is:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \qquad t = 1, \ldots, T. \tag{1}$$

where $A > 0$ is the amplitude, $L > 0$ the peak location, $S > 0$ the spread (scale), and $e(t)$ is additive noise. Given observations $(I(t))_{t=1}^{T}$, our goal is to estimate the underlying parameters using quantile regression, i.e., minimizing the asymmetric loss

$$\rho_\tau(u) = u\left(\tau - \mathbf{1}_{\{u<0\}}\right). \tag{2}$$

for a fixed quantile level $\tau \in (0, 1)$. We define the quantile objective:

$$f_\tau(T, a, l, s) = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau\left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right). \tag{3}$$

For fixed candidate values of (l, s), the optimal amplitude a minimizes $a \mapsto f_\tau(T, a, l, s)$. This minimizer is a weighted τ-quantile of a transformed version of the data, as described below.

2.1 Weighted Quantile Estimation of Amplitude

Let $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$ be the weight associated with time point t, and define:

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2}\right).$$

Then the optimal amplitude a is the weighted τ-quantile of $x(t)$ with weights $w(t)$. The following proposition gives the corresponding characterization of the weighted quantile as the minimizer of the weighted quantile loss. The proof of the following Proposition is the same as in the work of **?**.

Proposition 1. *Let $(x(t), w(t))$ be a sequence of real values with positive weights $w(t) > 0$. Then, for any $\tau \in (0, 1)$, the weighted τ-quantile minimizes the weighted quantile loss:*

$$q_\tau = \arg\min_{a \in \mathbb{R}} \sum_{t=1}^{T} w(t)\, \rho_\tau(a - x(t)),$$

where ρ_τ is the quantile check function defined in (2). To compute q_τ, sort the values $x(t)$ in increasing order, obtaining a permutation π such that

$$x(\pi(1)) \leq x(\pi(2)) \leq \cdots \leq x(\pi(T)).$$

Define the reordered weights $w_\pi(t) = w(\pi(t))$. Let

$$k = \min\left\{ j \in \{1, \ldots, T\} : \sum_{t=1}^{j} w_\pi(t) \geq \tau \sum_{t=1}^{T} w(t) \right\}.$$

Then the weighted τ-quantile satisfies $q_\tau = x(\pi(k))$.

2.2 Quantile-Based Estimation Procedure

We define the following steps to jointly estimate the location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and spread $\hat{s}(T)$:

Algorithm 1 Quantile-based estimation of peak parameters

Require: Time horizon T, upper bound $h > T$, quantile level $\tau \in (0, 1)$

1: For each candidate $l \in [1, h]$, find $s^*(l) = \arg\min_s f_\tau(T, a^*(l, s), l, s)$
2: Set $\hat{l}(T) = \arg\min_l f_\tau(T, a^*(l, s^*(l)), l, s^*(l))$
3: Set $\hat{s}(T) = s^*(\hat{l}(T))$
4: Set $\hat{a}(T) = a^*(\hat{l}(T), \hat{s}(T))$

Ensure: Estimated parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$

2.3 Remark on Asymmetry and Robustness

This method generalizes the classic Least Absolute Deviations (LAD) estimator, recovered at $\tau = 0.5$. By adjusting τ, we introduce asymmetry into the estimation loss, enabling the model to better adapt to skewed noise distributions. This makes the approach especially valuable in early-warning systems and epidemic peak detection where overestimation and underestimation have asymmetric consequences.

3 Quantile-Based Objective Function under Noise-Free Conditions

We first consider the idealized scenario where the signal is not corrupted by noise. The model of interest is given by

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right), \quad t = 1, \ldots, T,$$

where $A = 15000$, $L = 22$, $S = 10$, and $T \in \{2, 3, 10, 20, 21\}$. The quantile-based estimation procedure is implemented by minimizing the following objective function (3). The amplitude a is computed via a weighted τ-quantile, as described in Section 2.

Since the goal is to predict the location L of the peak before it becomes observable, we restrict the minimization over candidate peak locations $l > T$. For each value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute and visualize the objective curve $l \mapsto f_\tau(T, a(T, l), l, s(T, l))$ for the selected values of T (see Figure 1).

Remark. For each fixed sample size T, we computed the value of the minimized objective function at the estimated parameters $(\hat{a}, \hat{l}, \hat{s})$. The results indicate that the lowest objective value is typically achieved for $\tau = 0.95$, suggesting that upper quantiles align particularly well with the noiseless signal structure. The only exception is at $T = 20$, where $\tau = 0.75$ yields a slightly smaller loss. This empirical observation may help guide quantile selection when prior knowledge of signal shape or noise properties is limited.

4 Quantile-Based Estimation with τ-Aligned Noise

To analyze how quantile-aligned noise affects peak estimation, we introduce a unified noise model based on the *general asymmetric Laplace* (GAL) distribution. This flexible family allows both symmetric and asymmetric noise structures while maintaining the property that the τ-quantile of the noise is at zero.

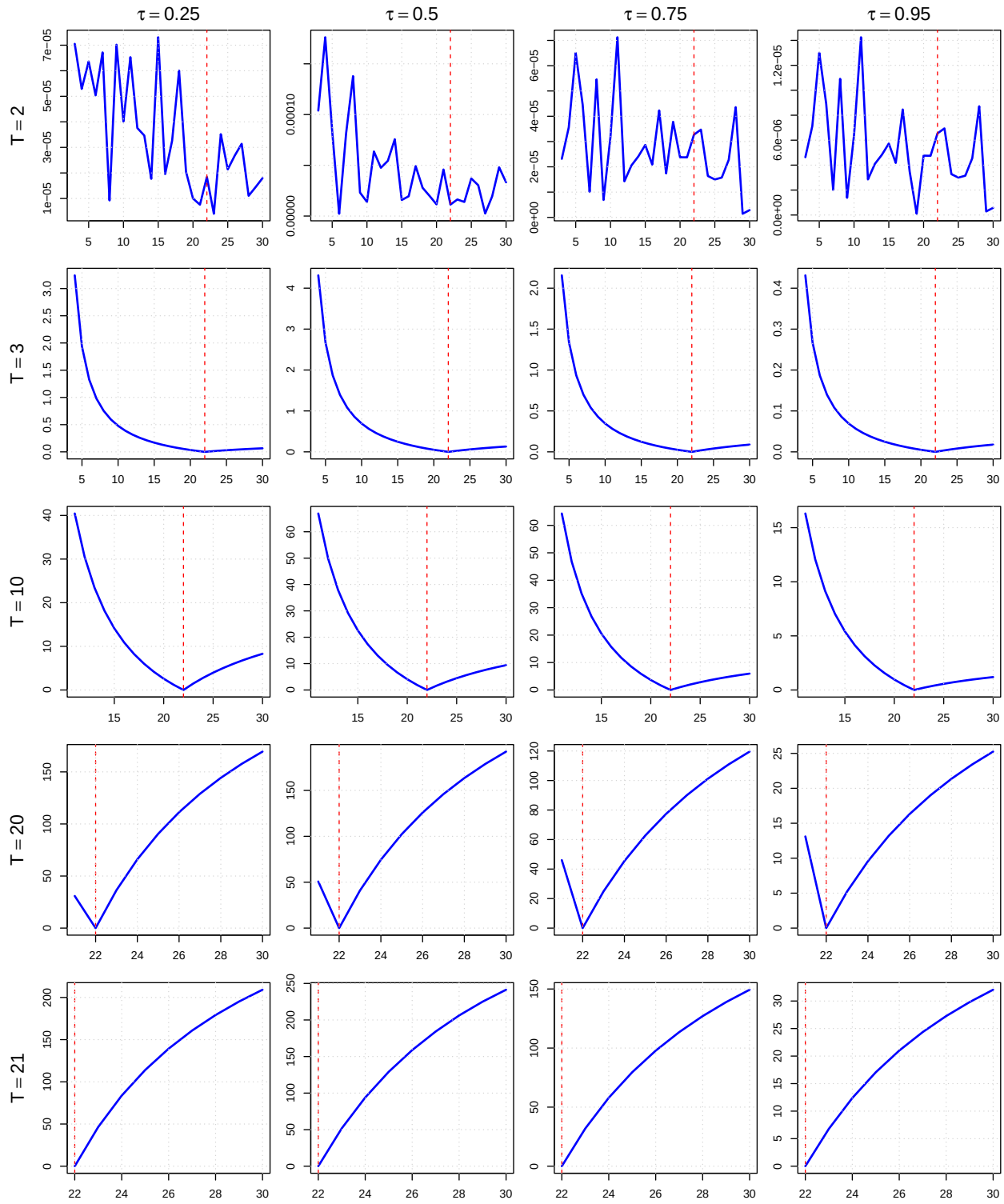


Figure 1: Objective function $l \mapsto f_\tau(T, a, l, s)$ computed for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{2, 3, 10, 20, 21\}$, under a noise-free signal. The vertical dashed line indicates the true peak location $L = 22$, and the search is restricted to values $l > T$.

4.1 Asymmetric Laplace Noise Aligned with the τ-Quantile

We consider an asymmetric Laplace distribution parametrized by a location parameter $m \in \mathbb{R}$, a scale parameter $\lambda > 0$, and an asymmetry parameter $k > 0$. Its density is given by

$$f_{m,\lambda,k}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda(x-m)}{k}\right), & x < m, \\ \exp\left(-\lambda k(x-m)\right), & x \geq m. \end{cases}$$

The parameter m is the mode of the distribution, while k controls its asymmetry and λ controls its concentration. The distribution function evaluated at the mode is

$$\mathbb{P}(X < m) = \frac{\lambda}{k + k^{-1}} \int_{-\infty}^{m} \exp\left(\frac{\lambda(x-m)}{k}\right) dx = \frac{k^2}{1+k^2}.$$

Therefore, to align the asymmetric Laplace noise with the τ-quantile regression criterion, we require

$$\mathbb{P}(e(t) < 0) = \tau, \tag{4}$$

and $\tau \in (0,1)$. Since the location of the noise can be fixed at zero, we set $m = 0$. The corresponding asymmetry parameter is then determined by

$$\frac{k^2}{1+k^2} = \tau,$$

or equivalently,

$$k = \sqrt{\frac{\tau}{1-\tau}}. \tag{5}$$

Thus, the τ-aligned asymmetric Laplace noise has density

$$f_{\tau,\lambda}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda x}{k}\right), & x < 0, \\ \exp(-\lambda k x), & x \geq 0, \end{cases}$$

where k is given by (5). By construction, the noise satisfies the τ-alignment property (4). In particular, when $\tau = 1/2$, we have $k = 1$, and the asymmetric Laplace distribution reduces to the symmetric Laplace distribution centered at zero. Hence, the symmetric Laplace case is recovered as a special case of the τ-aligned asymmetric Laplace model.

4.2 Simulation under Asymmetric Laplace Noise

We now investigate the behavior of the quantile-based peak estimation procedure when the additive noise follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1. In the parameterization based on the quantile loss, the density of the noise can be written as

$$f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b} \exp\left(-\frac{1}{b}\rho_\tau(x)\right), \qquad \tau \in (0,1), \quad b > 0. \tag{6}$$

where ρ_τ is the check function defined in (2). Equivalently,

$$f_{\tau,b}(x) = \begin{cases} \dfrac{\tau(1-\tau)}{b} \exp\left(-\dfrac{\tau x}{b}\right), & x \geq 0, \\ \dfrac{\tau(1-\tau)}{b} \exp\left(\dfrac{(1-\tau)x}{b}\right), & x < 0. \end{cases}$$

The normalization follows from

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{1}{b}\rho_\tau(x)\right) dx = \frac{b}{1-\tau} + \frac{b}{\tau} = \frac{b}{\tau(1-\tau)}.$$

Moreover, by construction, the noise satisfies the τ-alignment property (4). This distribution is directly compatible with the quantile-based estimation criterion. Indeed,

$$-\log f_{\tau,b}(e) = -\log\left(\frac{\tau(1-\tau)}{b}\right) + \frac{1}{b}\rho_\tau(e).$$

Therefore, for fixed τ and b, minimizing the negative log-likelihood is equivalent to minimizing the empirical quantile loss. The asymmetric Laplace model thus provides a probabilistic interpretation of the quantile-based peak estimation procedure without introducing a separate estimation criterion. For the simulation study, the true signal is

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right),$$

with $A = 1$, $L = 22$, $S = 10$. Since the maximum amplitude of the Gaussian signal is $\max_t I(t) = A$, we define the signal-to-noise ratio as the ratio between the peak amplitude of the signal and the standard deviation of the noise:

$$\mathrm{SNR} = \frac{A}{\sqrt{\mathrm{Var}(e(t))}}. \tag{7}$$

In the simulations, we set $\mathrm{SNR} = 1000$. For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the corresponding asymmetric Laplace noise is generated with asymmetry parameter (5). Thus, the noise variance is calibrated according to

$$\mathrm{Var}(e(t)) = \left(\frac{A}{\mathrm{SNR}}\right)^2. \tag{8}$$

For the τ-aligned asymmetric Laplace distribution, this variance is

$$\mathrm{Var}(e(t)) = b^2 \frac{\tau^2 + (1-\tau)^2}{\tau^2(1-\tau)^2}. \tag{9}$$

Combining (8) and (9) gives

$$b = \frac{A}{\mathrm{SNR}} \frac{\tau(1-\tau)}{\sqrt{\tau^2 + (1-\tau)^2}}. \tag{10}$$

For each τ, the objective function $l \mapsto f_\tau(T, a, l, s)$ is evaluated for the time horizons $T \in \{4, 6, 8, 9, 10\}$. For each candidate peak location l, the scale parameter s is optimized over the prescribed interval, and the amplitude a is determined as the weighted τ-quantile described in Section 2.1. The resulting objective profiles are then compared across the different values of τ. The resulting profiles are displayed in Figure 2. The true peak location $L = 22$ is marked by a vertical dashed line. Each panel corresponds to a different pair (T, τ), and the curves show the behavior of the quantile-loss objective across candidate peak locations.

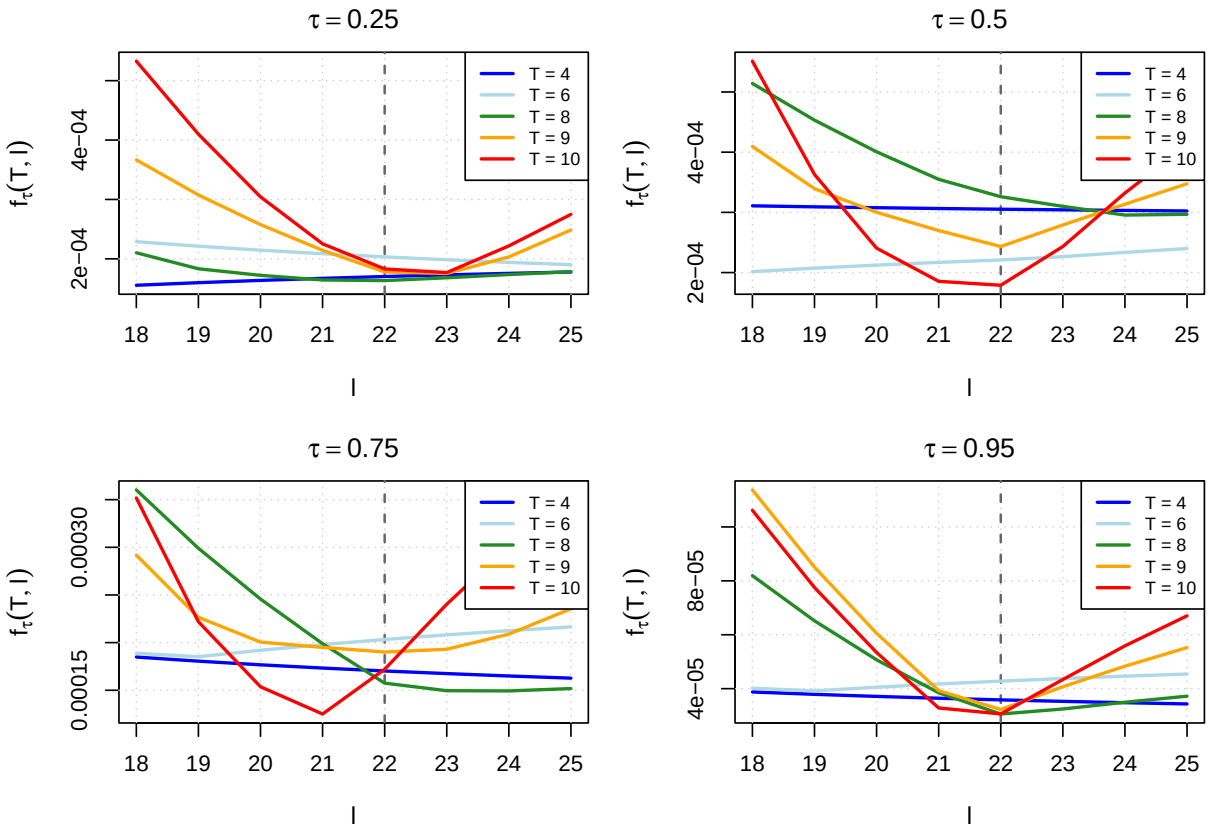


Figure 2: Objective function profiles $l \mapsto f_\tau(T, a, l, s)$ under *asymmetric Laplace* noise, for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$. Each subfigure corresponds to one value of τ, and the dashed vertical line indicates the true peak location $L = 22$.

5 Parameter Estimation Trajectories

We now analyze how the estimated model parameters, location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and scale $\hat{s}(T)$, evolve as the sample size T increases. Our goal is to assess the stability and convergence of these estimators under the τ-aligned asymmetric Laplace noise introduced in Section 4.1, including its symmetric special case. All experiments use the SNR definition given in (7), with SNR $= 1000$, and consider quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

5.1 τ-Aligned Asymmetric Laplace Noise

We consider the signal-plus-noise model (1), with $T = 30$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). The asymmetry parameter is determined by the quantile level according to (5), so that the τ-alignment property (4) holds. In particular, $\tau = 1/2$ corresponds to the symmetric Laplace distribution, whereas the other values of τ considered here produce asymmetric noise. The scale parameter b is calibrated separately for each τ so that the variance of the noise satisfies the fixed signal-to-noise ratio condition (8). For the τ-aligned asymmetric Laplace distribution, this gives (9), and therefore (10). At each time horizon $T \in \{4, \ldots, 30\}$, we estimate the parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ by minimizing the objective function (3) over the candidate peak locations and scale parameters, with the amplitude a re-estimated for each pair (l, s) as the weighted τ-quantile described in Section 2.1. The scale parameter s is optimized over the prescribed interval, and the location parameter l is selected by minimizing the resulting objective over the candidate locations. Figure 3 shows the resulting estimation trajectories $T \mapsto \hat{l}(T)$, $T \mapsto \hat{a}(T)$, $T \mapsto \hat{s}(T)$, for each quantile level τ. The true parameter values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

6 Stability Analysis

This section investigates the temporal stability of the estimated peak location when the signal-to-noise ratio is reduced. We focus on the τ-aligned asymmetric Laplace noise introduced in Section 4.1 and examine how the estimated peak location evolves as new observations become available. In particular, we compare the stability of the estimation paths across different quantile levels and assess the effect of a lower signal-to-noise ratio on the ability to recover the true peak location.

6.1 Low Signal-to-Noise Performance

To evaluate the robustness of the quantile-based estimator under a lower signal-to-noise ratio, we consider the signal-plus-noise model (1), with $T = 35$, $A = 1$, $L = 22$, $S = 10$. The noise $e(t)$ follows the τ-aligned asymmetric Laplace distribution introduced in Section 4.1, with density (6). For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, the scale parameter b is calibrated so that (8), with SNR $= 40$, according to the definition in (7). Using the variance of the τ-aligned asymmetric Laplace distribution gives (10). For each quantile level τ, we compute the estimation path $T \mapsto \hat{l}(T)$, $T = 4, \ldots, 35$, where $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ is obtained by minimizing the empirical quantile loss over the candidate peak locations and scale parameters, with the amplitude re-estimated as the weighted τ-quantile described in Section 2.1. To characterize the stability of the estimated peak location, we define an entry time as the earliest observation horizon T_{entry} such that all subsequent estimates remain within a prescribed tolerance of one

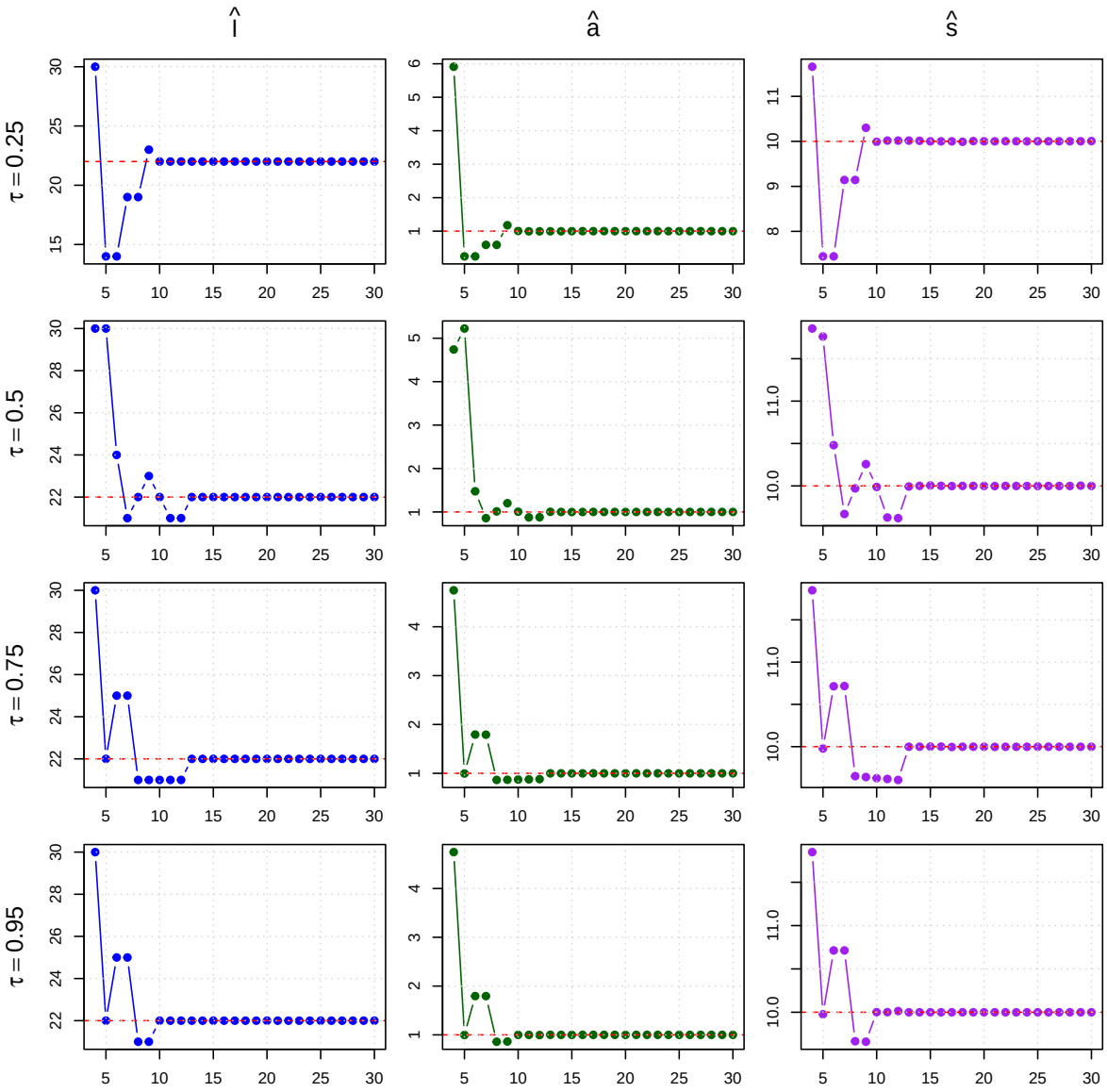


Figure 3: Estimation paths $T \mapsto \hat{l}(T)$ (left), $\hat{a}(T)$ (middle), and $\hat{s}(T)$ (right) under τ-aligned asymmetric Laplace noise with signal-to-noise ratio (SNR) 1000. Each row corresponds to a quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The true values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

another. Specifically, for a tolerance $\varepsilon = 4$, we define

$$T_{\text{entry}} = \min \left\{ T : \max_{u \geq T} \hat{l}(u) - \min_{u \geq T} \hat{l}(u) \leq \varepsilon \right\}.$$

This definition measures the earliest point from which the remaining estimation path remains confined to a stable range. Figure 4 displays the resulting estimation paths for the four quantile levels. The horizontal dashed line indicates the true peak location $L = 22$. The comparison illustrates how the temporal stability of the estimated peak location depends on the quantile level when the signal-to-noise ratio is reduced. The corresponding entry times are reported in Table 1. Smaller values of T_{entry} indicate that the estimator reaches a stable regime earlier, whereas larger values indicate that additional observations are required before the estimated peak location becomes stable.

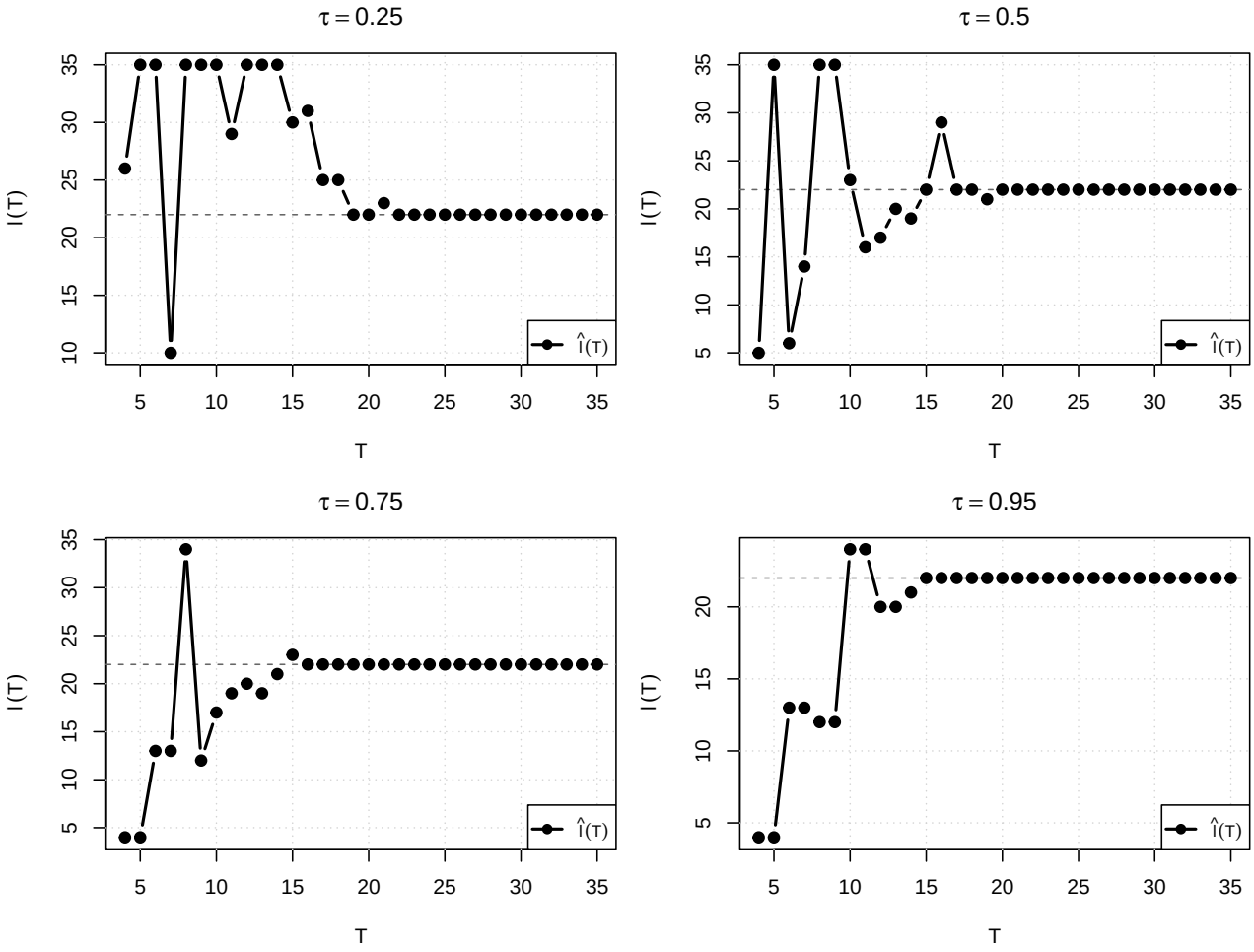


Figure 4: Estimation paths $T \mapsto \hat{l}(T)$ under a low signal-to-noise ratio (SNR $= 40$), for quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The horizontal dashed line indicates the true peak location $L = 22$.

Table 1: Entry times T_{entry} for the estimated peak location under SNR = 40 and different quantile levels.

	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.95$
T_{entry}	17	17	11	10

7 Entry and Exit Times of the Peak Region

The sequential estimation procedure introduced in the preceding sections provides, for each observation horizon T, an estimate $\hat{l}(T)$ of the peak location. The stabilization criterion considered in Section 6 characterizes the point from which the future estimation path remains within a prescribed tolerance. In this section, we introduce a complementary construction based on the oscillations of the estimated peak location. This construction aims at identifying a time interval describing the region in which the estimated peak location becomes stable, through a data-driven time of entry and a time of exit.

The procedure does not require knowledge of the true peak location L. The latter is used only in the simulation study, where it is known, to assess whether the estimated peak region contains the true peak.

7.1 Oscillations of the estimated peak location

Let

$$T_0 < T_0 + 1 < \cdots < h$$

denote the observation horizons for which the sequential estimator $\hat{l}(T)$ is available. For two observation times T and T' satisfying

$$T_0 \leq T < T' \leq h,$$

we define the oscillation of the estimated peak location over the interval $[T, T']$ by

$$\omega_l(T, T') = \max_{T \leq t_1, t_2 \leq T'} \left| \hat{l}(t_1) - \hat{l}(t_2) \right|. \tag{11}$$

Thus, $\omega_l(T, T')$ measures the largest variation of the estimated peak location over the interval $[T, T']$. In particular,

$$\omega_l(T, T') = 0$$

if and only if $\hat{l}(t)$ is constant over the observation horizons contained in $[T, T']$. A particular case of (11) is the future-tail oscillation

$$\omega_l(T, h) = \max_{T \leq t_1, t_2 \leq h} \left| \hat{l}(t_1) - \hat{l}(t_2) \right|. \tag{12}$$

This quantity measures the largest variation that can still occur in the estimated peak location after observation time T.

The future-tail oscillation is nonincreasing with T. Indeed, if

$$T_1 < T_2 \leq h,$$

then

$$[T_2, h] \subset [T_1, h],$$

and consequently

$$\omega_l(T_2, h) \leq \omega_l(T_1, h). \tag{13}$$

Hence, the function

$$g(T) = \omega_l(T, h)$$

provides a nonincreasing measure of the remaining temporal variation of the estimated peak location.

7.2 Candidate exit times

We use the evolution of the future-tail oscillation to construct candidate times of exit from the peak region. Let

$$g(T) = \omega_l(T, h).$$

A time T is said to correspond to a strict decrease of the future-tail oscillation if

$$g(T) > g(T + 1). \tag{14}$$

Let

$$T'_1 > T'_2 > \cdots > T'_K$$

denote the observation times corresponding to the strict decreases in (14), together with the terminal observation horizon h. For each such candidate, define

$$T^{(k)}_{\text{exit}} = T'_k \tag{15}$$

and

$$\tau'_k = \omega_l(T^{(k)}_{\text{exit}}, h). \tag{16}$$

The quantity τ'_k therefore represents the future-tail oscillation remaining at the candidate exit time $T^{(k)}_{\text{exit}}$. The corresponding candidate exit time is determined entirely from the observed estimation trajectory $\hat{l}(T)$ and does not require knowledge of the true peak location.

7.3 Determination of the entry time

For a given candidate exit time $T_{\text{exit}}^{(k)}$, we consider the oscillations accumulated between an earlier observation time T and the candidate exit time:

$$\omega_l(T, T_{\text{exit}}^{(k)}).$$

We first define the set

$$E_k = \left\{ T < T_{\text{exit}}^{(k)} : \omega_l(T, T_{\text{exit}}^{(k)}) \geq \tau_k' \right\}. \tag{17}$$

If E_k is nonempty, let

$$T_k^* = \max E_k. \tag{18}$$

We then define the corresponding oscillation level by

$$\tau_k = \omega_l(T_k^*, T_{\text{exit}}^{(k)}). \tag{19}$$

The entry time associated with the candidate exit time is the first observation horizon at which this oscillation level is attained:

$$T_{\text{entry}}^{(k)} = \min \left\{ T < T_{\text{exit}}^{(k)} : \omega_l(T, T_{\text{exit}}^{(k)}) = \tau_k \right\}, \tag{20}$$

provided that the minimum is attained.

The resulting candidate peak region is therefore the open interval

$$\mathcal{R}_k = \left(T_{\text{entry}}^{(k)}, T_{\text{exit}}^{(k)} \right), \tag{21}$$

with corresponding width

$$w_k = T_{\text{exit}}^{(k)} - T_{\text{entry}}^{(k)}. \tag{22}$$

This construction provides an interval rather than a single stabilization time. The left endpoint represents the beginning of the selected stable region, while the right endpoint represents the corresponding candidate exit time identified from the future-tail oscillation.

7.4 Interpretation and simulation assessment

The construction above is entirely based on the sequential estimates $\hat{l}(T)$. In particular, the true peak location L is not used in determining either $T_{\text{entry}}^{(k)}$ or $T_{\text{exit}}^{(k)}$. This is essential for applications in which the true peak location is unknown.

In simulations, however, the true peak location L is known and can therefore be used for an ex post assessment of the estimated peak region. We say that the kth candidate region contains the true peak if

$$T_{\text{entry}}^{(k)} < L < T_{\text{exit}}^{(k)}. \tag{23}$$

This criterion is used only for evaluating the procedure and is not involved in its construc-

tion.

When several candidate regions satisfy (23), their widths can be compared through (22). A narrower interval provides a more localized description of the time region associated with the estimated peak, while still containing the true peak.

The proposed procedure can therefore be summarized as follows:

1. compute the sequential peak-location estimates $\hat{l}(T)$;
2. compute the future-tail oscillation $\omega_l(T, h)$;
3. identify candidate exit times from the strict decreases of $\omega_l(T, h)$;
4. for each candidate exit time, compute the local oscillations $\omega_l(T, T_{\text{exit}}^{(k)})$;
5. determine the corresponding entry time from (20);
6. construct the candidate peak region $\mathcal{R}_k$;
7. in simulations, assess whether the region contains the true peak using (23).

The resulting entry and exit times provide a complementary characterization of temporal stability. Whereas the stabilization criterion of Section 6 identifies a single observation horizon after which the estimated peak location remains within a prescribed tolerance, the present construction produces an explicit interval associated with the evolution of the estimator's oscillations.

7.5 Simulation study

We illustrate the entry–exit procedure using the same Gaussian peak model and asymmetric Laplace noise framework considered in the simulation study. In particular, the sequential estimates are obtained by minimizing the quantile-loss criterion for the selected quantile level τ, with the noise scale calibrated according to the signal-to-noise ratio definition given in Section **??**.

For a fixed simulation setting, we first compute the trajectory

$$T \longmapsto \hat{l}(T),$$

followed by the future-tail oscillation

$$T \longmapsto \omega_l(T, h).$$

The candidate exit times are then obtained from the strict decreases of the latter curve. For each candidate exit time, the corresponding entry time and interval width are computed according to the procedure above.

The simulation results are reported in terms of the candidate quadruples

$$\left(\tau_k, \tau_k', T_{\text{entry}}^{(k)}, T_{\text{exit}}^{(k)}\right),$$

together with the corresponding interval widths w_k. Since the true peak location is known in the simulation setting, we additionally report whether

$$T_{\text{entry}}^{(k)} < L < T_{\text{exit}}^{(k)}.$$

A graphical illustration of the procedure will display the trajectory $T \mapsto \hat{l}(T)$ together with the future-tail oscillation $T \mapsto \omega_l(T, h)$ and the selected entry–exit interval. The effect of the signal-to-noise ratio on the resulting peak region will also be investigated.